01 · THE PROBLEM

HELP IS 22 MINUTES AWAY

On Mars, a radio signal takes about **3 to 22 minutes** to reach Earth, one way. Asking a flight surgeon a question and hearing back can take the better part of an hour, and when the Sun sits between the planets, contact can drop out for about two weeks.



A crew may not include a doctor. When someone gets sick, they have to decide on their own: **is this routine, or is it an emergency?**

02 · OUR GOAL

Help a crew member with no medical training assess symptoms and know how urgently to act, on board, with no link to Earth.

A **Fully offline.** No internet, no cloud.

B **Grounded.** Every answer is tied to a cited medical source.

C **Repeatable.** The same symptoms always get the same answer.

D **No fake data.** Sensor slots stay empty until real hardware fills them.

THE CHALLENGE

NASA HUNCH 2026–27

Software & Technology: **Medical Diagnostic System with Machine Learning & Artificial Intelligence**

TEAM TETHER

Joshua Collado
Project lead · main coder, diagnostic engine

Joaquin
Medical research · rules · co-coder

Cruz
UI design · user testing

Instructor: Ms. Sylvia Odoms
Mentor: Michael Hayes, Voyager Technologies

NASA HUNCH 2026–27 · TEAM TETHER

VITALS

An offline AI medical triage assistant for crews too far from Earth to call a doctor

03 HOW IT WORKS

1
Crew describes symptoms
In plain words, typed in: “I’ve had a headache and felt dizzy since this morning.”

2
Extract findings
The description is turned into structured medical findings, and negations like “no fever” are caught.

3
Retrieve conditions
Only the conditions that match are pulled from our cited medical knowledge base.

4
Read vital signs
Heart rate, SpO_2, blood pressure and temperature from on-board sensors (hardware in progress).

5
Local AI reasons
A language model running in Ollama on an NVIDIA Jetson weighs the evidence. No internet needed.

6
Answer + escalation
Likely conditions, an urgency level, the next question to ask, and citations from the knowledge base.

04 GUARDRAILS IN THE CODE

01 The AI can only name conditions from the knowledge base. Anything it invents is dropped and reported.

02 It never writes a citation. Sources are looked up from the knowledge base.

03 Temperature 0: the same complaint always gets the same answer, so it can be tested.

04 It tracks what is ALREADY KNOWN and STILL UNKNOWN, so it never re-asks a question.

05 If the AI names a serious condition but forgets to escalate, the code escalates anyway.

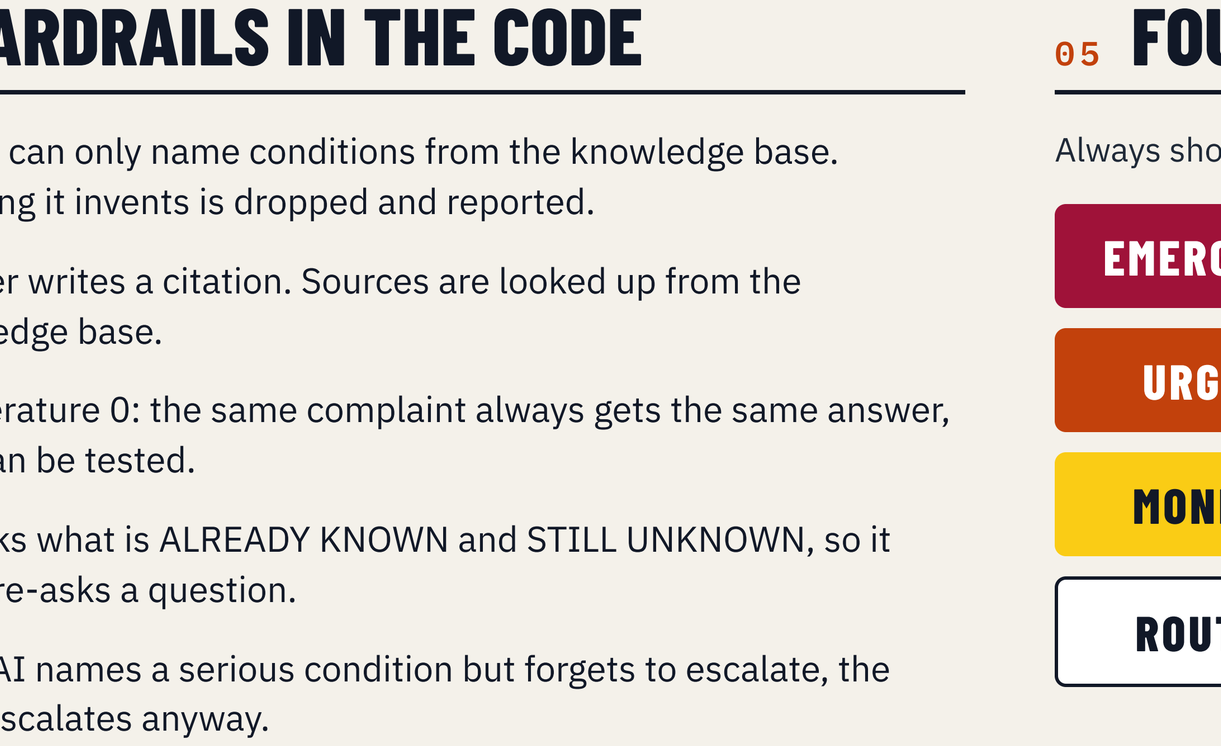
05 FOUR URGENCY LEVELS

Always shown as a word *and* a color, never color alone.

EMERGENCY	Act immediately	2 conditions
URGENT	Needs prompt attention	2 conditions
MONITOR	Watch and reassess	5 conditions
ROUTINE	Handle with normal care	5 conditions

THE ASTRONAUT SCREEN

Demo case · flight day 63



14
medical conditions in the knowledge base

72
findings it can recognize

0
internet connection needed

06 · HARDWARE

SENSORS & COMPUTER

The AI runs on an **NVIDIA Jetson** inside Docker, with the model built into the image so it starts with no download. Sensors are planned on a Raspberry Pi.

Heart rate + SpO_2

SELECTED

MAX30102 pulse oximeter: reads the pulse from light through the fingertip

Blood pressure

IN PROGRESS

Salvaged cuff, micro pump and valve, a 0–40 kPa pressure sensor, and a MOSFET drive circuit

Temperature

EVALUATING

Comparing contact and non-contact sensors

Respiration

OPTIONAL

60 GHz mmWave radar, a no-touch option

07 · INTERFACE

BUILT FOR THE CREW

- Name and password login, plus a red Emergency button that skips it
- A side rail: chats, profile, vitals, doctor notes and supplies
- The urgency level comes first, as a word and a color
- One follow-up question at a time, and a “Wrong?” button to fix a misread
- 44 px touch targets, 4.5:1 contrast, runs fully offline

08 · TESTING

BUGS WE CAUGHT BY RUNNING IT

- “No fever” leaked into the next sentence → fixed with clause-boundary checks
- Re-asked questions already answered → added the ALREADY KNOWN list
- One red flag set off five alarms → added escalation gates

Next: a nurse, EMT or doctor will review the knowledge base’s condition weights before we rely on them.

09 · TIMELINE

SEP 1, 2026
Project outline

NOV 2026
Preliminary Design Review

MAR 2027
Critical Design Review

MAY 4–6, 2027
Final Design Review · Johnson Space Center

DATA SOURCES

NASA Human Research Roadmap: In-flight Medical Conditions evidence report · NASA Open Science Data Repository (OSDR) · Life Sciences Data Archive (LSDA)

Scan for the website

blackphenixlord.github.io/vitals-site

github.com/Blackphenixlord/lunar-medical-diagnostic